

Aravind Reddy Talla

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EDUCATION

Northwestern University	Evanston, Illinois, USA
Ph.D. in Computer Science, Advisors: Konstantin Makarychev and Aravindan Vijayaraghavan	June 2023
M.S. in Computer Science, GPA: 4.0/4.0	March 2021
Indian Institute of Technology (IIT) Kanpur	
Bachelor of Technology (B.Tech.) in CSE with minor in Physics, Distinction	May 2018

EMPLOYMENT

IIT Madras	
Principal Project Scientist, Centre for Responsible AI (CeRAI), Wadhvani School of Data Science & AI	3 Nov 2025 -
IIT Hyderabad	
Visiting Assistant Professor, Department of Artificial Intelligence (AI)	22 July 2024 - 21 July 2025
Broad Institute of MIT and Harvard	Cambridge, Massachusetts, USA
Postdoctoral Associate, Mentor: Victoria Popic	05 July 2023 - 10 Jan 2024
Adobe Research	San Jose, California, USA (remote)
Intern, Primary mentor: Ritwik Sinha	Summer 2022
Intern, Primary mentors: Ryan Rossi and Zhao Song	Summer 2021
Paris Centre for Quantum Technologies, Université Paris Diderot	Paris, France
Intern, Mentors: Eleni Diamanti and Iordanis Kerenidis, partly funded by Charnak Scholarship	Summer 2017
Centre for Quantum Technologies, National University of Singapore	Singapore
Intern, Mentor: Hartmut Klauck	Summer 2016

REFEREED INTERNATIONAL CONFERENCE PROCEEDINGS

7. DiRe: Diversity-promoting Regularization for Dataset Condensation	
with Saumyanjan Mohanty and Konda Reddy Mopuri, conference proceedings link	WACV 2026
6. Online Adaptive Mahalanobis Distance Estimation	
with Lianke Qin and Zhao Song, conference proceedings link	BigData 2023
5. Dynamic Tensor Product Regression	
with Zhao Song and Lichen Zhang, conference proceedings link	NeurIPS 2022
4. One-Pass Algorithms for MAP Inference of Nonsymmetric Determinantal Point Processes	
with Ryan Rossi, Zhao Song, and others, conference proceedings link	ICML 2022
3. Adaptive and Dynamic Multi-Resolution Hashing for Pairwise Summations	
with Lianke Qin, Zhao Song, Zhaozhuo Xu, and Danyang Zhuo, conference proceedings link	BigData 2022
2. Beyond Perturbation Stability: LP Recovery Guarantees for MAP Inference on Noisy Stable Instances	
with Hunter Lang, David Sontag, and Aravindan Vijayaraghavan, conference proceedings link	AISTATS 2021
1. Improved Guarantees for k-means++ and k-means++ Parallel	
with Konstantin Makarychev and Liren Shan, conference proceedings link	NeurIPS 2020

GRANTED US PATENT

1. Online Inference and Learning for Nonsymmetric Determinantal Point Processes	2025
with Ryan Rossi, Zhao Song, and others, Google Patents link	US12288237B2

AWARDS & RESEARCH GRANTS

Adobe Research Gift Grant (\$20,000) , awarded for a grant application I wrote to support my PhD	2022
Peter and Adrienne Barris Outstanding Teaching Assistant Award , Northwestern, press release	2022
CS Supplemental Award (\$3,000) , Northwestern, for being one of the top CS PhD applicants	2018
Charpak Scholarship for Research Internship , Government of France	2017
Academic Excellence Award , IIT Kanpur, for being in the top 10% of the batch by first-year CPI	2015
National Initiative on Undergraduate Science Scholarship , HBCSE (TIFR)	2015
All India Rank 189 , JEE Advanced 2014	2014
Gold Medallist , Indian National Physics Olympiad (INPhO) 2014, <i>National top 35</i>	2014

CONFERENCE TRAVEL GRANTS

NeurIPS 2022 Scholar Award (\$2000)	2022
FOCS 2022 Travel Award (\$425)	2022
ICML 2022 Participation Grant (\$400)	2022
STOC 2019 Student Travel Grant (\$600)	2019

TEACHING EXPERIENCE

Instructor at IIT Hyderabad:

- Convex Optimization/Optimization-1 (AI 2101/AI 1233) : Full semester course, Jan - May 2025, Course Rating: 4.03/5.00, Instructor rating: 4.02/5.00 (158 students enrolled, 124 ratings submitted)
- Programming for AI (AI 1013): 10 week course, co-taught with Dr. P.N. Karthik, Course Rating: 4.71/5.00, Instructor rating (only for me): 4.58/5.00 (36 students enrolled, 34 ratings submitted)
- Introduction to Modern Artificial Intelligence (AI 1001) : 5 week course, Sep - Oct 2024, Course rating: 4.07/5.00, Instructor rating: 4.02/5.00, (103 students enrolled, 90 ratings submitted)
- Introduction to Submodular Functions (EE 5328) : 5 week course, July - Sep 2024, Course rating: 4.35/5.00, Instructor rating: 4.70/5.00, (5 students enrolled, 4 ratings submitted)

Teaching Assistant at Northwestern:

- Mathematical Foundations of Computer Science (CS 212) - Spring 2022, Fall 2020, Fall 2019
- Received an award for being the **best TA** in the Spring 2022 quarter, [press release](#).
- Design & Analysis of Algorithms (CS 336) - Winter 2019

Teaching Assistant at IIT Kanpur: Fundamentals of Computing (ESC 101) - Winter 2018

Volunteer Math Tutor: As a part of the non-profit [Tutoring Chicago](#) program, I tutored a Chicago area middle-school student for the 2021-22 academic year, mainly focusing on middle-school and high-school math.

Guest lectures at Northwestern:

- Submodular Function Maximization Algorithms for Big Data, April 2022
- Introduction to Markov Chains Math Toolkit for Theoretical Computer Scientists, May 2019

SERVICE AND OUTREACH

Conference Reviewer:

- AAAI Conference on Artificial Intelligence (AAAI) 2026
- International Conference on Artificial Intelligence and Statistics (AISTATS) 2023 - 2024
- European Conference on Computer Vision (ECCV) 2026
- Foundations of Software Technology and Theoretical Computer Science (FSTTCS) 2024
- International Conference on Learning Representations (ICLR) 2022 - 2024

- International Conference on Machine Learning (ICML) 2022 - 2026
* **Silver Reviewer Award**, ICML 2026 Position Paper Track
- Neural Information Processing Systems (NeurIPS) 2021 - 2025

Invited Speaker: For 300 high-school students, Oak Valley International School, Hyderabad, Jun 2025

Workshop Leader and Mentor: NeurIPS 2022 High-School Outreach Program

Board Member: CS PhD Advisory Council (NU, 2020 - 2022)

Webmaster: Northwestern CS Theory group (2018 - 2023)

Coordinator at IIT Kanpur student groups: Association for Computing Activities (ACA), Card & Board Games Club (CBG), and Science Coffeehouse (SCH) (2016 - 17)

TECHNICAL SKILLS

Programming Languages : Python, C

Natural Languages : English, Telugu, Hindi - Native/Bilingual Proficiency;
Korean - Limited Working Proficiency

INVITED TALKS AND POSTER SESSIONS

Beyond Worst-Case Analysis of Algorithms in Data Science & AI:

- IIT Madras, Department of Data Science and Artificial Intelligence (DSAI) Apr 2025

On MAP Inference of Ferromagnetic Potts Models and Nonsymmetric Determinantal Point Processes :

- IIT Madras, Department of Data Science and Artificial Intelligence (DSAI) Sep 2024
- IIT Kanpur, Department of Computer Science and Engineering (CSE) Jun 2024
- IISc Bangalore, as a part of Bangalore Theory Seminar, YouTube link May 2024
- IIT Hyderabad, Department of CSE and Department of AI May 2024

Dynamic Tensor Product Regression :

- Neural Information Processing Systems (NeurIPS) Dec 2022, New Orleans, Louisiana, USA

On Clustering, MAP Inference, and Causal Inference :

- Broad Institute of MIT and Harvard, Popic Lab Nov 2022, Cambridge, Massachusetts, USA
- Stanford University, Aghaeepour Lab Oct 2022, Stanford, California, USA

Beyond Perturbation Stability: LP recovery guarantees for MAP Inference on noisy stable instances :

- University of Chicago Theory Lunch Nov 2022, Chicago, Illinois, USA
- International Conference on Artificial Intelligence and Statistics (AISTATS) April 2021, virtual

One-Pass Algorithms for MAP Inference of Nonsymmetric Determinantal Point Processes :

- NSF TRIPODS meeting (poster) Sep 2022, virtual
- International Conference on Machine Learning (ICML) July 2022, Baltimore, Maryland, USA

Improved Guarantees for k-means++ and k-means++ Parallel :

- Neural Information Processing Systems (NeurIPS) Dec 2020, virtual